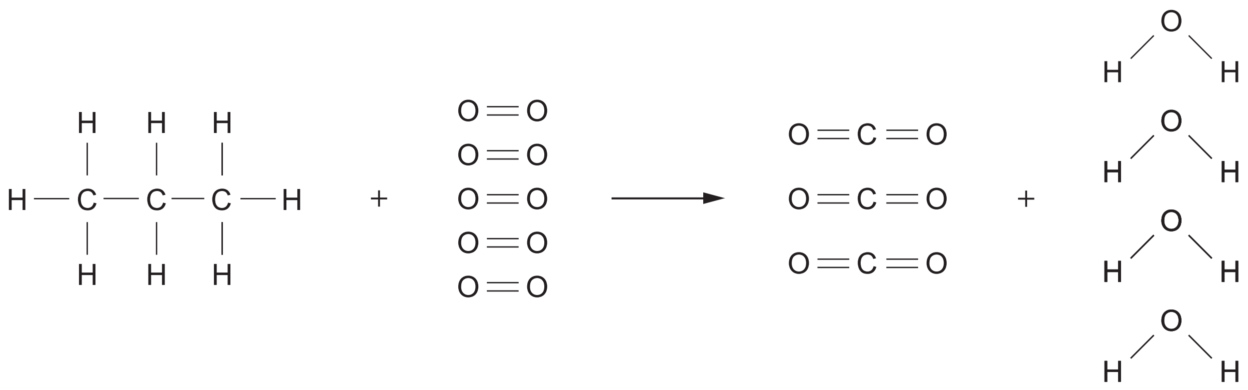
Examiner only

3. (a) Propane gas is commonly used as a fuel. The burning of propane gas is represented by the following equation.



The relative amounts of energy needed to break some of the bonds are given in the table.

Bond	Bond energy (kJ)
C—H	413
C—C	348

(i) The total energy needed to break all of the bonds in the reactants is 6490 kJ.

Calculate the energy needed to break **one** O=O bond. [3]

Energy needed = ... kJ

 (3430UE0-1)

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(ii) The energy released when all of the bonds in the products are formed is 8 542 kJ.

Tick (✓) the correct overall energy change for this reaction. [1]

2 052 kJ taken in	☐
15 032 kJ taken in	☐
15 032 kJ given out	☐
2 052 kJ given out	☐

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(b) Propane gas is commonly used in camping stoves.

The photograph shows a camping stove being used to heat 750 g of water.

When heated for 10 minutes, 220 kJ of energy was transferred to the water.

Use the following equation to calculate the temperature rise of the water. [1]

$$\text{temperature rise (°C)} = \frac{\text{energy transferred (J)}}{4.2 \times \text{mass of water (g)}}$$

Temperature rise = ... °C

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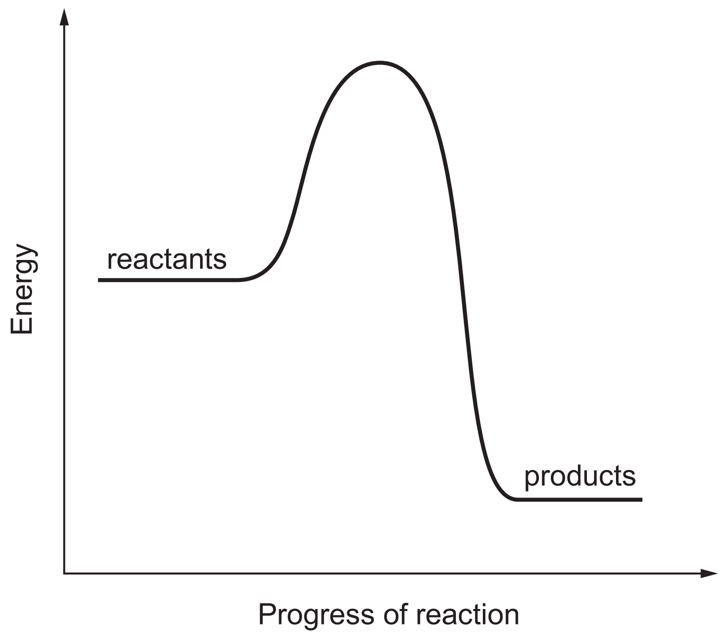
(c) The energy profile diagram for the combustion of propane is given below.



(i) Draw an **arrow** (↕) **on the diagram** to show the activation energy.

Label this **A**. [1]

(ii) Draw an **arrow** (↕) **on the diagram** to show the overall energy change.

Label this **B**. [1]

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